Analysis-0 2026 Spring

Homework 7

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*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

$\boxed{\textbf{Problem 1} \text{ (4-2).}}$ Let f be a continuous mapping from a metric space X into a metric space Y. Prove that for every set $E \subset X$,

$$f(\overline{E}) \subset \overline{f(E)},$$

where $\overline{E}$ denotes the closure of E. Give an example showing that $f(\overline{E})$ can be a proper subset of $\overline{f(E)}$.

Proof. Take $p \in \overline{E}$. We will show that $f(p) \in \overline{f(E)}$, i.e. for any $\varepsilon > 0$, there exists $q \in E$ such that $d(f(p), f(q)) < \varepsilon$.

Since f is continuous at p, there exists $\delta > 0$ such that for any $x \in X$ with $d(p, x) < \delta$, we have $d(f(p), f(x)) < \varepsilon$. Since $p \in \overline{E}$, there exists $q \in E$ such that $d(p, q) < \delta$. Thus, we have $d(f(p), f(q)) < \varepsilon$, which completes the proof of $f(\overline{E}) \subset \overline{f(E)}$.

For the second part, let $X = Y = \mathbb{R}$ with the metric $d(x, y) = |x - y|$, and $f(x) = \arctan x$. Then, for $E = \mathbb{R}$, we have $f(E) = f(\overline{E}) = (-\pi/2, \pi/2)$ but $\overline{f(E)} = [-\pi/2, \pi/2]$, which shows that $f(\overline{E})$ is a proper subset of $\overline{f(E)}$. $\square$

$\boxed{\textbf{Problem 2} \text{ (4-4).}}$ Let f and g be continuous mappings from a metric space X into a metric space Y, and let E be a dense subset of X.

Prove that $f(E)$ is dense in $f(X)$. If $g(p) = f(p)$ for every $p \in E$, prove that $g(p) = f(p)$ for every $p \in X$. (In other words, a continuous mapping is determined by its values on a dense subset of its domain.)

Proof. By the previous problem, we have $f(\overline{E}) \subset \overline{f(E)}$. Since E is dense in X, we have $\overline{E} = X$, which implies that $f(X) \subset \overline{f(E)}$. Thus, $f(E)$ is dense in $f(X)$.

For the second part, let

$$h: \ X \to \mathbb{R}, \ x \to d(f(x), g(x)).$$

It's not hard to verify that h is continuous on X and $h(p) = 0$ for every $p \in E$. Since E is dense in X, we have $\{0\}$ is dense in $h(X)$. Since $h(X)$ is a subset of $\mathbb{R}$, we have $h(X) = \{0\}$, which implies that $f(p) = g(p)$ for every $p \in X$. $\square$

Problem 3 (4-6). If f is defined on E, then the graph of f is the set of points $(x, f(x))$ with $x \in E$. In particular, if E is a subset of the real numbers and f is real-valued, then the graph of f is a subset of the plane.

Assume E is compact. Prove that f is continuous on E if and only if its graph is compact.

Proof. Denote $g(x) = (x, f(x))$ as a function from E to $\mathbb{R}^2$, and define $S = g(E)$. Then g is continuous if f is continuous. Assume that $E \times f(E)$ is equipped with the product topology.

(1) "$\Longrightarrow$". For any open cover $\{G_\alpha\}$ of S, we have $\tilde{G}_\alpha := g^{-1}(G_\alpha \cap S)$ is an open cover of E. Since E is compact, there exists a finite subcover $\{\tilde{G}_{\alpha_i}\}$ of E. Then, $\{G_{\alpha_i}\}$ is a finite subcover of S, which implies that S is compact.

(2) "$\Longleftarrow$". For any $x \in E$ and $\varepsilon > 0$, we will prove that there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for any $y \in N_\delta(x)$.

Denote

$$G = N_\varepsilon(x) \times N_\varepsilon(f(x)), \; G_n = \big(E - \overline{N}_{\varepsilon/n}(x)\big) \times f(E),$$

where $\overline{N}$ denote the closed neighborhood. Then, by the definition of product topology, G and G_n is open in $E \times f(E)$.

Since for every $y \neq x$, we have $(y, f(y)) \in G_n$ for sufficiently large n, and $(x, f(x)) \in G$, therefore $\{G\} \cup \{G_n : \; n \in \mathbb{N}\}$ is an open cover of S. Since S is compact, there exists a finite subcover $\{G\} \cup \{G_{n_i} : i = 1, 2, \cdots, k\}$ of S. Let $N = \max\{n_i : \; i = 1, 2, \cdots, k\}$. Then, for any $y \in E$ with $|x - y| < \varepsilon/N$, we have $(y, f(y)) \in G$, which implies that $|f(x) - f(y)| < \varepsilon$.

Take $\delta = \varepsilon/N$, we have $|f(x) - f(y)| < \varepsilon$ for any $y \in N_\delta(x)$, which completes the proof. □

Problem 4 (4-8). Let f be a uniformly continuous real-valued function on a bounded set $E \subset \mathbb{R}^1$. Prove that f is bounded on E.

If the boundedness assumption on E is removed, show that this conclusion need not hold.

Proof. Since $f(E)$ is not bounded, there exists a sequence $\{x_n\} \in E$ such thath $|f(x_{n+1})| > 1 + |f(x_n)|$ for all $n \geq 1$. Since E is bounded, there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that $x_{n_k} \to x_0$ for some $x_0 \in \mathbb{R}$. Since f is uniformly continuous on E, then for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in E$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$. Take k_0 such that $|x_{n_k} - x_0| < \delta/2$ for any $k \geq k_0$. Then, for any $k, l \geq k_0$, we have

$$|f(x_{n_k}) - f(x_{n_l})| < \varepsilon,$$

which contradicts the fact that $|f(x_{n+m})| > m + |f(x_n)|$ for all $m, n \geq 1$.

For the second part, let $E = \mathbb{R}$ and $f(x) = x$. Then, f is uniformly continuous on E but $f(E)$ is not bounded. □

Problem 5 (4-9). Prove that the requirement in the definition of uniform continuity can be expressed in terms of diameters as follows: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $E \subset X$ with $\operatorname{diam}(E) < \delta$, one has $\operatorname{diam}(f(E)) < \varepsilon$.

Proof. First, we will show that the original definition of uniform continuity implies the diameter version. For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in X$ with $d(x, y) < \delta$, we have $d(f(x), f(y)) < \varepsilon$. For any $E \subset X$ with $\operatorname{diam}(E) < \delta$, we have $d(x, y) < \delta$ for any $x, y \in E$. Thus, we have $d(f(x), f(y)) < \varepsilon$ for any $x, y \in E$, which implies that $\operatorname{diam}(f(E)) < \varepsilon$.

Next, we will show that the diameter version implies the original definition of uniform continuity. For any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $E \subset X$ with $\operatorname{diam}(E) < \delta$, we have $\operatorname{diam}(f(E)) < \varepsilon$. For any $x, y \in X$ with $d(x, y) < \delta$, we have $\operatorname{diam}(\{x, y\}) = d(x, y) < \delta$. Thus, we have $\operatorname{diam}(f(\{x, y\})) < \varepsilon$, which implies that $d(f(x), f(y)) < \varepsilon$.

This completes the proof. □

Problem 6 (4-10). Supply the details of the following alternative proof of Theorem 4.19:

If f is not uniformly continuous, then there exists $\varepsilon > 0$ and two sequences $\{p_n\}, \{q_n\}$ in X such that

$$d_X(p_n, q_n) \to 0,$$

but

$$d_Y\big(f(p_n), f(q_n)\big) > \varepsilon$$

for all n. Use Theorem 2.37 to derive a contradiction.

Proof. Suppose that f is not uniformly continuous. Then, there exists $\varepsilon > 0$ such that for any $\delta > 0$, there exist $p, q \in X$ with $d(p, q) < \delta$ but $d(f(p), f(q)) > \varepsilon$. For each $n \in \mathbb{N}$, take $p_n, q_n \in X$ such that $d(p_n, q_n) < 1/n$ but $d(f(p_n), f(q_n)) > \varepsilon$. Then, we have $d(p_n, q_n) \to 0$ but $d(f(p_n), f(q_n)) > \varepsilon$ for all n.

Since $\{p_n\}$ lies in a compact metric space X, there exists a subsequence $\{p_{n_k}\}$ of $\{p_n\}$ such that $p_{n_k} \to p_0$ for some $p_0 \in X$. Since $d(p_{n_k}, q_{n_k}) < 1/n_k$, we have $q_{n_k} \to p_0$ as well. Since f is continuous at p_0, we have $f(p_{n_k}) \to f(p_0)$ and $f(q_{n_k}) \to f(p_0)$. Thus, we have

$$d(f(p_{n_k}), f(q_{n_k})) \to 0,$$

which contradicts the fact that $d(f(p_{n_k}), f(q_{n_k})) > \varepsilon$ for all k.

This completes the proof. □

Problem 7 (4-12). The composition of uniformly continuous functions is uniformly continuous.

State this assertion precisely, and prove it.

Proof. Let X, Y, Z be metric spaces, and $f: X \to Y$, $g: Y \to Z$ be uniformly continuous functions. We will show that the composition $g \circ f: X \to Z$ is uniformly continuous.

For any $\varepsilon > 0$, since g is uniformly continuous on Y, there exists $\delta_1 > 0$ such that for any $y_1, y_2 \in Y$ with $d(y_1, y_2) < \delta_1$, we have $d(g(y_1), g(y_2)) < \varepsilon$. Since f is uniformly continuous on X, there exists $\delta_2 > 0$ such that for any $x_1, x_2 \in X$ with $d(x_1, x_2) < \delta_2$, we have $d(f(x_1), f(x_2)) < \delta_1$. Thus, for any $x_1, x_2 \in X$ with $d(x_1, x_2) < \delta_2$, we have

$$d(g(f(x_1)), g(f(x_2))) < \varepsilon,$$

which implies that $g \circ f: X \to Z$ is uniformly continuous.

This completes the proof. □

Problem 8 (4-15). Let f be a mapping of X into Y. If for every open set $V \subset X$, the image $f(V)$ is open in Y, then f is called an open mapping.

Prove that every continuous open mapping of $\mathbb{R}^1$ into $\mathbb{R}^1$ is monotone.

Proof. Suppose that f is not monotone. Then, there exist $a, b, c \in \mathbb{R}$ with $a < c < b$ such that $f(c) > \max\{f(a), f(b)\}$ or $f(c) < \min\{f(a), f(b)\}$. Without loss of generality, we assume that $f(c) > \max\{f(a), f(b)\}$.

Let $\eta = \sup\{f(x) : a < x < b\}$. Since f is continuous on (a, b), and $\eta \geq f(c) > \max\{f(a), f(b)\}$, there exist $\xi \in (a, b)$ such that $f(\xi) = \eta$.

Let $\eta' = \inf\{f(x) : a < x < b\}$. For every $y \in (\eta', \eta]$, there exists $x \in (a, b)$ such that $\eta' < f(x) < y < f(\xi) = \eta$, therefore by the continuity of f, there exists $\xi_y \in (a, b)$ such that $f(\xi_y) = y$. Thus, we have $(\eta', \eta] \subset f((a, b))$, hence

$$f((a, b)) = (\eta', \eta] \text{ or } [\eta', \eta],$$

which is a contradiction since f is an open mapping and (a, b) is open in $\mathbb{R}^1$.

This completes the proof. □

Problem 9 (4-18). Every rational number x can be written in the form $x = m/n$, where $n > 0$ and m, n are integers with no common factor. When $x = 0$, take $n = 1$. Consider the function f on $\mathbb{R}^1$ defined by

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational,} \\ \dfrac{1}{n}, & \text{if } x = \dfrac{m}{n}. \end{cases}$$

Prove that f is continuous at every irrational point, and that at every rational point f has a simple discontinuity.

Proof. (1) For any irrational point x_0, we will show that f is continuous at x_0.

For any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $1/N < \varepsilon$. Since the set of rational numbers with denominator less than or equal to N is finite, there exists $\delta > 0$ such that for any rational number m/n with $n \leq N$, we have $d(x_0, m/n) \geq \delta$.

Thus, for any $x \in \mathbb{R}$ with $d(x, x_0) < \delta$, we have either x is irrational and $f(x) = 0 = f(x_0)$, or $x = m/n$ is rational with $n > N$ and $f(x) = 1/n < 1/N < \varepsilon$. In either case, we have $d(f(x), f(x_0)) < \varepsilon$, which implies that f is continuous at every irrational point.

(2) For any rational point $x_0 = m_0/n_0$, we will show that f has a simple discontinuity at x_0. If f is continous at $x_0 \in \mathbb{Q}$, then for any sequence $\{x_n\}$ with $x_n \to x_0$, we have $f(x_n) \to f(x_0) = 1/n_0$.

However, we can take a sequence $\{x_n\}$ of irrational numbers such that $x_n \to x_0$, which implies that $f(x_n) = 0 \to 0 \neq 1/n_0$. Thus, f is not continuous at any rational point.

This completes the proof. □

Problem 10 (4-20). Let E be a nonempty subset of a metric space X. Define the distance from $x \in X$ to E by

$$\rho_E(x) = \inf_{z \in E} d(x, z).$$

(a) Prove that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

(b) For all $x, y \in X$, prove that

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y).$$

Then use this to prove that ρ_E is uniformly continuous on X.

Proof. (a) If $x \in \overline{E}$, then for any $\varepsilon > 0$, there exists $z \in E$ such that $d(x, z) < \varepsilon$. Thus $\rho_E(x) = \inf_{z \in E} d(x, z) \leq d(x, z) < \varepsilon$. Since ε is arbitrary, we have $\rho_E(x) = 0$.

Conversely, if $\rho_E(x) = 0$, then for any $\varepsilon > 0$, there exists $z \in E$ such that $d(x, z) < \varepsilon$. Thus, $x \in \overline{E}$.

(b) For any $x, y \in X$, we have

$$|\rho_E(x) - \rho_E(y)| = \left| \inf_{z_1 \in E} d(x, z_1) - \inf_{z_2 \in E} d(y, z_2) \right| \leq \inf_{z \in E} |d(x, z) - d(y, z)| \leq d(x, y).$$

□

Problem 11 (4-23). A real-valued function f on (a, b) is called convex if, whenever $a < x < b$, $a < y < b$, and $0 < \lambda < 1$,

$$f\big(\lambda x + (1 - \lambda) y\big) \leq \lambda f(x) + (1 - \lambda) f(y).$$

Prove that every convex function is continuous. Then prove that every increasing convex function of a convex function is convex. (For example, if f is convex, then e^f is also convex.)

If f is convex on (a, b) and $a < s < t < u < b$, prove that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t}.$$

Proof. We will prove the inequality first. For any $a < s < t < u < b$, we have

$$t = \lambda s + (1 - \lambda) u,$$

where $\lambda = (u - t)/(u - s)$. Since f is convex, we have

$$f(t) \leq \lambda f(s) + (1 - \lambda) f(u),$$

which implies that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s}.$$

Similarly, we have

$$f(t) \leq \lambda' u + (1 - \lambda') s,$$

where $\lambda' = (t-s)/(u-s)$. Thus, we have

$$\frac{f(u)-f(s)}{u-s} \le \frac{f(u)-f(t)}{u-t}.$$

This completes the proof of the inequality.

Next, we will show that f is continuous. For any $a < x_0 < b$, select $a_0, b_0 \in (a,b)$ such that $a_0 < x_0 < b_0$. Then, for any $x \in (x_0, b_0)$, we have

$$\frac{f(x_0)-f(a_0)}{x_0-a_0} \le \frac{f(x)-f(x_0)}{x-x_0} \implies f(x) \ge f(x_0) + \frac{f(x_0)-f(a_0)}{x_0-a_0}(x-x_0),$$
$$\frac{f(x)-f(x_0)}{x-x_0} \le \frac{f(b_0)-f(x_0)}{b_0-x_0} \implies f(x) \le f(x_0) + \frac{f(b_0)-f(x_0)}{b_0-x_0}(x-x_0).$$

The same argument and conclusion holds for $x \in (a_0, x_0)$. Then, we have

$$\begin{aligned} f(x_0) = \lim_{x\to x_0}\left(f(x_0) + \frac{f(b_0)-f(x_0)}{b_0-x_0}(x-x_0)\right) \\ \le \lim_{x\to x_0} f(x) \\ \le \lim_{x\to x_0}\left(f(x_0) + \frac{f(x_0)-f(a_0)}{x_0-a_0}(x-x_0)\right) = f(x_0), \end{aligned}$$

which implies $\lim_{x\to x_0} f(x)$ exists and equals $f(x_0)$. Thus, f is continuous at x_0. Since x_0 is arbitrary, we have f is continuous on (a,b).

Finally, we will show that if $g : \mathbb{R} \to \mathbb{R}$ is an increasing convex function and $f : (a,b) \to \mathbb{R}$ is a convex function, then the composition $g \circ f : (a,b) \to \mathbb{R}$ is convex.

For any $a < x, y < b$ and $0 < \lambda < 1$, we have

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y).$$

Since g is increasing, we have

$$g(f(\lambda x + (1-\lambda)y)) \le g(\lambda f(x) + (1-\lambda) f(y)).$$

Since g is convex, we have

$$g(\lambda f(x) + (1-\lambda) f(y)) \le \lambda g(f(x)) + (1-\lambda) g(f(y)).$$

Thus, we have

$$g(f(\lambda x + (1-\lambda)y)) \le \lambda g(f(x)) + (1-\lambda) g(f(y)),$$

which implies that $g \circ f : (a,b) \to \mathbb{R}$ is convex. □

Problem 12 (4-26). Let X, Y, and Z be metric spaces, with Y compact. Let f map X into Y, let g be a one-to-one continuous mapping of Y into Z, and for each $x \in X$ define $h(x) = g\big(f(x)\big)$.

Prove that if h is uniformly continuous, then f is uniformly continuous.

Proof. Without loss of generality, we assume that $Z = g(Y)$. Then $f = g^{-1} \circ h$. Hence, it suffices to show that $g^{-1} : Z \to Y$ is uniformly continuous.

Since g is continuous and Y is compact, we have $Z = g(Y)$ is compact as well. For any closed set $F \in Y$, F is compact in Y, hence $g(F)$ is compact in Z. Since Z is a metric space, $g(F)$ is closed in Z. Thus, g^{-1} is continuous. Since Z is compact, we have g^{-1} : is uniformly continuous. □

Problem 13 (5-7). Assume that $f'(x)$ and $g'(x)$ exist, $g'(x) \neq 0$, and $f(x) = g(x) = 0$. Prove that

$$\lim_{t \to x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}.$$

(This also holds for complex-valued functions.)

Proof. By the definition of derivative, we have

$$f'(x) = \lim_{t \to x} \frac{f(t)}{t - x},\ g'(x) = \lim_{t \to x} \frac{g(t)}{t - x}.$$

Then, for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $t \in \mathbb{R}$ with $0 < |t - x| < \delta$, we have

$$\left| \frac{f(t)}{t - x} - f'(x) \right| < \varepsilon,\ \left| \frac{g(t)}{t - x} - g'(x) \right| < \varepsilon.$$

Thus, for any $t \in \mathbb{R}$ with $0 < |t - x| < \delta$,

$$\begin{aligned}
\left| \frac{f(t)}{g(t)} - \frac{f'(x)}{g'(x)} \right| &= \left| \frac{f(t)}{t - x} \frac{t - x}{g(t)} - \frac{f'(x)}{g'(x)} \right| \\
&\leq \left| \frac{f(t)}{t - x} - f'(x) \right| \left| \frac{t - x}{g(t)} \right| + |f'(x)| \cdot \left| \frac{t - x}{g(t)} - \frac{1}{g'(x)} \right| \\
&< \varepsilon \left| \frac{t - x}{g(t)} \right| + |f'(x)| \left| \frac{g(t)}{t - x} - g'(x) \right| \left| \frac{t - x}{g(t) g'(t)} \right| \\
&< \varepsilon \left(g'(x) + \varepsilon \right) + \varepsilon \left| f'(x) \right| \frac{1}{(|g'(x)| - \varepsilon) g'(x)}.
\end{aligned}$$

Since ε is arbitrary, we have

$$\lim_{t \to x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}.$$

This completes the proof. □

Problem 14 (5-13). Let $a, c \in \mathbb{R}$ with $c > 0$, and let f be defined on $[-1, 1]$ by

$$f(x) = \begin{cases} |x|^a \sin\left(|x|^{-c}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Prove the following statements:

(a) f is continuous if and only if $a > 0$.

(b) $f'(0)$ exists if and only if $a > 1$.

(c) f' is bounded if and only if $a \geq 1 + c$.

(d) f' is continuous if and only if $a > 1 + c$.

Proof. (a) If $a > 0$, then for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in \mathbb{R}$ with $|x| < \delta$, we have $|f(x)| \leq |x|^a < \varepsilon$. Thus, f is continuous at 0. Since f is continuous at every nonzero point, we have f is continuous on $[-1, 1]$.

Conversely, if $a \leq 0$, then there exists a sequence $\{t_n\}$ with $t_n \to 0^+$ such that $|t_n|^{-c} = (4n+1)\pi/2$ for all $n \geq 1$. Thus, we have $f(t_n) = |t_n|^a \sin(|t_n|^{-c}) = |t_n|^a \to 1$ or ∞, which implies that f is not continuous at 0.

(b) If $a > 1$, then for any $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x \in \mathbb{R}$ with $|x| < \delta$, we have

$$\left| \frac{f(x) - f(0)}{x - 0} - 0 \right| = \left| \frac{f(x)}{x} \right| \leq |x|^{a-1} < \varepsilon.$$

Thus, $f'(0) = 0$ exists.

Conversely, if $a \leq 1$, then take $\{t_n\}$ as in the proof of (a), and $\{s_n\}$ such that $s_n \to 0^+$ and $|s_n|^{-c} = (2n)\pi$ for all $n \geq 1$. Then, we have

$$\left| \frac{f(t_n)}{t_n} \right| = |t_n|^{a-1} \to 1 \text{ or } \infty, \quad \left| \frac{f(s_n)}{s_n} \right| = 0$$

, which implies that $f'(0)$ does not exist.

(c) For $x > 0$, we have

$$f'(x) = ax^{a-1} \sin(x^{-c}) - cx^{a-c-1} \cos(x^{-c}).$$

If $a \geq 1 + c > 1$, then $|ax^{a-1} \sin(x^{-c})| \leq a\,|x|^{a-1}$ is bounded on $(0, 1]$, and $|cx^{a-c-1} \cos(x^{-c})| \leq cx^{a-c-1}$ is also bounded on $(0, 1]$. Thus, f' is bounded on $(0, 1]$. Since f' is bounded on $[-1, 0)$ as well, we have f' is bounded on $[-1, 1]$.

Conversely, if $a < 1 + c$, then take $\{s_n\}$ as in the proof of (b). Then, we have

$$f'(s_n) = -cs_n^{a-c-1} \to -\infty,$$

which implies that f' is not bounded on $[-1, 1]$.

(d) If $a > 1 + c$, then $f'(0) = 0$ exists by (b). Then,

$$\left|ax^{a-1} \sin(x^{-c})\right| = a\,|x|^{a-1} \to 0, \ \left|cx^{a-c-1} \cos(x^{-c})\right| = cx^{a-c-1} \to 0 \ (x \to 0).$$

Hence $\lim_{x \to 0} f'(x) = 0$, therefore f' is continuous at 0. Since f' is continuous at every nonzero point, we have f' is continuous on $[-1, 1]$.

Conversely, if $a \leq 1 + c$, then take $\{s_n\}$ as in the proof of (b). Then, we have

$$f'(s_n) = -cs_n^{a-c-1} \to -c \text{ or } -\infty,$$

which implies that f' is not continuous at 0. $\square$